Hema Sri Sai Kollipara

: kollipa4@msu.edu| : +1-517-974-8557 | : www.linkedin.com/in/hemasrisai-k/
Work Authorization: F-1 OPT (STEM-eligible)

Professional Summary

Analytics Consultant/Data Scientist with strong experience in applying machine learning, statistical modeling, and simulation techniques to solve real-world business problems. Proven track record of translating complex data into actionable insights that support decision-making, performance optimization, and strategic planning. Experienced in end-to-end analytics delivery including problem formulation, data preparation, model development, validation, and stakeholder communication. Proficient in Python, R, SQL, and dashboarding tools, with experience working across academic, business, and executive stakeholders.

Education

- **Michigan State University (Ph.D. in Statistics)** — Michigan, USA
  Focus: Applied Machine Learning, Predictive Modeling, Bayesian Methods — *Aug 2019 - Aug 2025*
- **University of Hyderabad (Integrated Masters in Mathematics)** — Hyderabad, India
  Major in Statistics — *Aug 2011 - May 2016*

Data Science & Analytics Experience

Graduate Teaching Assistant & Researcher (Michigan State University) — Aug 2019 - Aug 2025

- Designed and validated predictive models for high-dimensional datasets, focusing on improving model reliability, interpretability, and decision accuracy in complex real-world settings.
- Built and benchmarked machine learning and statistical models using simulation-based validation to assess robustness, generalization, and performance trade-offs across methods.
- Developed end-to-end modeling pipelines in Python and R, covering data preprocessing, feature selection, model estimation, and performance evaluation.
- Applied advanced Bayesian and regularization techniques to improve signal extraction and reduce overfitting in large-scale prediction problems.
- Collaborated across research teams to translate analytical results into actionable insights, ensuring clarity for both technical and non-technical stakeholders.
- Maintained reproducible and scalable workflows using Git, Quarto, R Markdown, and Shiny to support collaborative development and reporting.

Research Assistant (Indian School of Business, Hyderabad) — July 2016 - July 2019

- Developed and deployed machine learning models to analyze customer behavior and predict outcomes, improving prediction accuracy by 19% and directly supporting profitability initiatives.
- Designed simulation-based and optimization models to evaluate advertising performance and budget allocation strategies, achieving a 70% increase in audience engagement efficiency.
- Created and delivered applied analytics content and case studies for 300+ professionals in executive education programs, bridging theory with real business applications.
- Applied natural language processing techniques to extract sentiment and performance insights from unstructured text data, linking trends to audience engagement metrics.
- Collaborated closely with faculty and stakeholders on data collection, modeling, and interpretation to ensure analytical outputs aligned with business and organizational objectives.
- Delivered a Power BI dashboard analyzing institutional performance, directly supporting ISB's advancement in the Financial Times Global MBA ranking.

Key Projects

Predictive Risk Scoring System:

- Developed credit-default risk models for customers without formal credit history using machine learning and Bayesian neural networks.
- Performed feature engineering, model tuning, and benchmarking against traditional logistic regression and random forest baselines.
- Achieved a 10% improvement in predictive accuracy, enabling better risk differentiation and decision support for lending strategies. (Tools: Python, PyTorch, scikit-learn)

Advertising Reach Optimization:

- Built simulation-based models to quantify the relationship between advertising reach and Gross Rating Points (GRPs) across budget scenarios.
- Identified saturation thresholds and diminishing returns, enabling optimized budget allocation strategies.
- Improved audience reach efficiency by 70% and reduced redundant ad spend. (Tools: R, tidyverse)

Executive Performance Dashboard:

- Designed and delivered an interactive Power BI dashboard for executive stakeholders to monitor institutional performance, admissions, and placement metrics.
- Insights supported strategic planning and coincided with an improvement in global MBA rankings. (Tools: Power BI, Excel)

Scalable Dimensionality Reduction for Count Data:

- Developed a scalable analytics algorithm for high-dimensional, over-dispersed datasets.
- Improved computational efficiency by 20%, enabling faster analysis and model experimentation. (Tools: R, C++, Rcpp)

Applied Modeling & Analytics Approach

- Frame analytical problems by first identifying decision objectives, operational constraints, and success metrics before selecting modeling techniques.
- Emphasize simulation-based validation, stress testing, and robustness checks to ensure models generalize beyond training data and perform reliably in practice.
- Balance predictive performance with interpretability to support stakeholder understanding and adoption of model-driven insights.
- Design modeling pipelines that are reproducible, scalable, and adaptable to evolving data, requirements, and performance feedback.

Applied Analytics & AI Focus Areas

Predictive modeling • Performance optimization • Simulation-based validation • Decision support systems • Dashboard-driven insights • High-dimensional data modeling • Model interpretability and robustness

Certifications

- **Data Analyst Associate Certificate : DataCamp, 2024** Hands-on certification demonstrating applied SQL, Python, and data visualization skills through real-world data cleaning, querying, and KPI analysis projects.
- **AI Engineer for Data Scientists Associate : DeepLearning.AI (Coursera), 2025** Specialized in neural network deployment, model optimization, and machine learning pipelines using TensorFlow and Keras.
- **Machine Learning : Stanford University (Coursera), 2025** Completed a comprehensive course covering supervised learning, clustering, anomaly detection using reinforcement learning and evaluation metrics using real-world datasets.

Technical Skills

- **Programming & Data:** SQL, R, Python,MATLAB.
- **Machine Learning & AI:** Predictive Modeling, Neural Networks, NLP, Feature Engineering, Model Evaluation
- **Statistical Methods:** Bayesian Modeling, Regularization, Simulation-Based Validation
- **Data Analysis & Visualization:** Power BI, Tableau, Matplotlib, KPI Dashboards
- **Big Data & Cloud Technologies:** Exposure to cloud and big-data platforms (AWS, Spark, Databricks)

Publications

- K., H.S.S., Pal, M., & Manimaran, P. (2019). Multifractal detrended partial cross-correlation analysis on Asian markets. Physica A: Statistical Mechanics and its Applications, 531, 121778. doi.org/10.1016/j.physa.2019.121778
- Pochiraju, B., & Kollipara, H. S. S. (2019). Statistical Methods: Regression Analysis (Chapter-7). In Essentials of Business Analytics (pp. 179-245). Springer, Cham. doi.org/10.1007/978-3-319-68837-4$_7$
- Agrawal, D., Kollipara, H. S. S., & Mamidipudi, S. (2019). Case Study: AAA Airline. In Essentials of Business Analytics (pp. 863-871). Springer, Cham. dx.doi.org/10.1007/978-3-319-68837-4$_2$6
- Kollipara, Hema Sri Sai., Tapabrata Maiti, Sanjukta Chakraborty, and Samiran Sinha. ”Benchmarking Sparse Variable Selection Methods for Genomic Data Analyses.” Statistics in Medicine 45, no. 3-5 (2026): e70428. doi.org/10.1002/sim.70428.